

FMO



Case: Expected Returns

- Suppose you have predicted the following returns for stocks Cefal and Toten in three possible states of nature. What are the expected returns?

○ State	Probability	Cefal	Toten
○ Boom	0.3	0.15	0.25
○ Normal	0.5	0.10	0.20
○ Recession	???	0.02	0.01

- $R_C = .3(.15) + .5(.10) + .2(.02) = .099 = 9.99\%$
- $R_T = .3(.25) + .5(.20) + .2(.01) = .177 = 17.7\%$

Variance and Standard Deviation

- Variance and standard deviation still measure the volatility of returns
- Using unequal probabilities for the entire range of possibilities
- Weighted average of squared deviations
- It's important to point out that these formulas are for populations
- Standard deviation is the square root of the variance

$$\sigma^2 = \sum_{i=1}^n p_i (R_i - E(R))^2$$

Example: Variance and Standard Deviation

- Consider the previous example. What are the variance and standard deviation for each stock?
- Stock Cefal
 - $\sigma^2 = .3(.15-.099)^2 + .5(.1-.099)^2 + .2(.02-.099)^2 = .002029$
 - $\sigma = .045$
- Stock Toten
 - $\sigma^2 = .3(.25-.177)^2 + .5(.2-.177)^2 + .2(.01-.177)^2 = .007441$
 - $\sigma = .0863$

Another Example

- Consider the following information:

State	Probability	ABC, Inc.
◦ Boom	.25	.15
◦ Normal	.50	.08
◦ Slowdown	.15	.04
◦ Recession	.10	-.03

- What is the expected return?
- What is the variance?
- What is the standard deviation?

Another Example

$$E(R) = .25(.15) + .5(.08) + .15(.04) + .1(-.03) = .0805$$

$$\text{Variance} = .25(.15-.0805)^2 + .5(.08-.0805)^2 + .15(.04-.0805)^2 + .1(-.03-.0805)^2 = .00267475$$

$$\text{Standard Deviation} = .051717985$$

More specifically, variance measures how far each number in the set is from the mean (average)



Return



The variability of returns from those that are expected.

What rate of return do you expect on your investment (savings) this year?

What rate will you actually earn?

Does it matter if it is a bank CD or a share of stock?



Return



How to Determine the Expected Return and Standard Deviation

Stock BW			The expected return, $\bar{R}$, for Stock BW is .09 or 9%
R_i	P_i	$(R_i)(P_i)$	
-0.15	0.10	-0.015	
-0.03	0.20	-0.006	
0.09	0.40	0.036	
0.21	0.20	0.042	
0.33	0.10	0.033	
Sum	1.00	0.090	



Determining Standard Deviation (Risk Measure)

$$\sigma = \sqrt{\sum_{i=1}^n (R_i - \bar{R})^2 (P_i)}$$

 **Standard Deviation, σ** , is a statistical measure of the variability of a distribution around its mean.
It is the square root of variance.
Note, this is for a *discrete distribution*.



Standard Deviation

$$\sigma = \sqrt{\sum_{i=1}^n (R_i - \bar{R})^2 (P_i)}$$

$$\sigma = \sqrt{.01728}$$

$$\sigma = 0.1315 \text{ or } 13.15\%$$

Coefficient of variation

The ratio of the *standard deviation* of a distribution to the *mean* of that distribution.

It is a measure of *RELATIVE* risk.

$$CV = \sigma / \bar{R}$$

$$CV \text{ of BW} = 0.1315 / 0.09 = 1.46$$



Case



<https://www.icicidirect.com/ilearn/stocks/courses/chapter-6-risk-reward-matrix-for-investment>

- Home
- iLearn
- Personal Finance
- Courses
- Chapter 6: Risk Reward Matrix For Investment



Risk Attitude

Certainty equivalent $>$ Expected value
Risk Preference

Certainty equivalent = Expected value
Risk Indifference

Certainty equivalent $<$ Expected value
Risk Aversion

Most individuals are Risk Averse.



Risk Attitude Case

You have the choice between (1) a guaranteed dollar reward or (2) a coin-flip gamble of \$100,000 (50% chance) or \$0 (50% chance). The expected value of the gamble is **\$50,000**

- *Mary* requires a guaranteed \$25,000, or more, to call off the gamble.
- *Rony* is just as happy to take \$50,000 or take the risky gamble.
- *Shannon* requires at least \$52,000 to call off the gamble.



Risk Attitude Case

What are the Risk Attitude tendencies of each?

Mary shows “**risk aversion**” because her “certainty equivalent” $<$ the expected value of the gamble.

Rony exhibits “**risk indifference**” because her “certainty equivalent” equals the expected value of the gamble.

Shannon reveals a “**risk preference**” because her “certainty equivalent” $>$ the expected value of the gamble.





Total Risk = Systematic Risk + Unsystematic Risk

$$\text{Total Risk} = \text{Systematic Risk} + \text{Unsystematic Risk}$$

Systematic Risk is the variability of return on stocks or portfolios associated with changes in return on the market as a whole.

Unsystematic Risk is the variability of return on stocks or portfolios not explained by general market movements. It is avoidable through diversification.





Unsystematic Risk

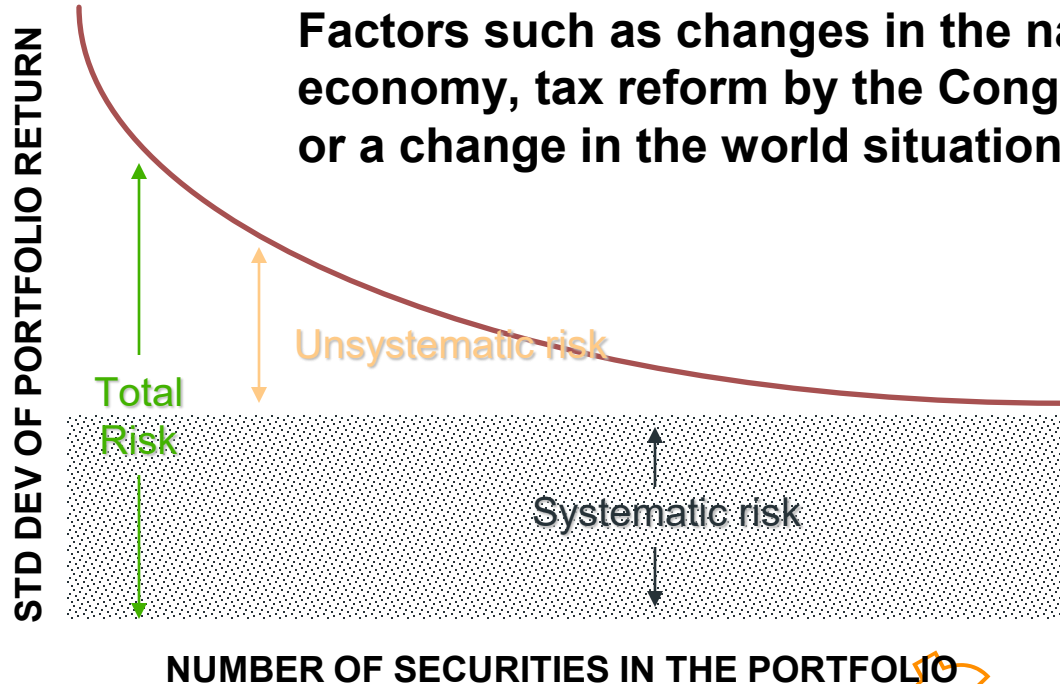
Risk factors that affect a limited number of assets

Also known as unique risk and asset-specific risk

Includes such things as labor strikes, part shortages, etc.



Total Risk = Systematic Risk + Unsystematic Risk





Diversifiable Risk

The risk that can be eliminated by combining assets into a portfolio

Often considered the same as unsystematic, unique or asset-specific risk

If we hold only one asset, or assets in the same industry, then we are exposing ourselves to risk that we could diversify away





Systematic Risk Principle

- There is a reward for bearing risk
- There is not a reward for bearing risk unnecessarily
- The expected return on a risky asset depends only on that asset's systematic risk since unsystematic risk can be diversified away





Systematic Risk Principle

We use the beta coefficient to measure systematic risk
What does beta tell us?

- A beta of 1 implies the asset has the same systematic risk as the overall market
- A beta < 1 implies the asset has less systematic risk than the overall market
- A beta > 1 implies the asset has more systematic risk than the overall market





Beta and the Risk Premium

Remember that the risk premium = expected return – risk-free rate

The higher the beta, the greater the risk premium should be

Can we define the relationship between the risk premium and beta so that we can estimate the expected return?

YES!



Portfolio Expected Returns and Betas

